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PROBLEM SET 5

1. Greedy Algorithm

```
def Greedy_Money(denoms, cents):
# denoms → array of denomination of change
to be used
# cents → change required
    num_coins = 0
    for i in range(0, len(denoms)):
        while cents ≥ denoms[i]:
            num_coins = num_coins + 1
            cents = cents - denoms[i]
return num_coins
```

a) 87 cents

\# coins returned = 6

optimal solution returned

b) 80 cents

\# coins returned = 8

<u>NOT</u> optimal solution

returning 2 quaters & 3 dimes

would return 5 coins

2. $\boxed{\log_b(xy) = \log_b x + \log_b y}$

step #	Action	Justification
1	let $x = b^M$ $\therefore \log_b x = M$	premise (log rule)
2	let $y = b^N$ $\therefore \log_b y = N$	premise (log rule)
3	$\log_b(xy)$	premise
4	$\log_b(b^M \cdot b^N)$	substitute step 1 & 2 → 3
5	$\log_b(b^{M+N})$	properties of exponents
6	$M + N$	properties of logs $\log_b(b^c) = c$
7	$\log_b x + \log_b y$	sub 1 & 2 → 6

$\boxed{\log_b(x^y) = y \cdot \log_b x}$

Step	Action	Justification
1	let $x = b^M$	premise
2	$\log_b(x) = M$	log rule
3	$\log_b(x^y)$	premise
4	$\log_b((b^M)^y)$	Sub. 1→3
5	$\log_b(b^{M \cdot y})$	prop. exponents
6	$M \cdot y$	prop. of logs
7	$\log_b(x) \cdot y$	Sub. 2→6
8	$y \cdot \log_b(x)$	commutative multiplication

2. continued

$$\boxed{\log_a(x) = \frac{\log_b(x)}{\log_b(a)}}$$

Step #	Action	Justification
1	let $\log_a(x) = M$	premise
2	$a^M = x$	prop. logs
3	$\log_b(a^M) = \log_b(x)$	$\log_b$ of both sides
4	$M \cdot \log_b(a) = \log_b(x)$	prop logs (see prev. page)
5	$M = \frac{\log_b(x)}{\log_b(a)}$	solve for m
6	$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$	sub $1 \to 5$

3.

Step #	Action	Justification
1	$n < 2^n$	premise
2	$\log_2(n) < \log_2(2^n)$	take log base 2 of both sides
3	$\log(n) < n$	simplify 2
4	let $c = 1$ & $k = 1$ then $(1) \cdot \log(1) < 1$	find witness
5	$0 < 1$	$\log(n)$ is $O(n)$ because n is greater than $\log(n)$

5. a) $A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}$ $B = \begin{bmatrix} b_{1,1} & b_{2,1} \\ b_{2,1} & b_{2,2} \end{bmatrix}$ $C = \begin{bmatrix} c_{1,1} \\ c_{1,2} \end{bmatrix}$

$m \times n$ $n \times k$ $k \times p$

2×2 2×2 2×1

$$AB = \begin{bmatrix} a_{1,1} \cdot \overset{①}{b_{1,1}} + a_{1,2} \cdot \overset{②}{b_{2,1}} & a_{1,1} \cdot \overset{⑤}{b_{2,1}} + a_{1,2} \cdot \overset{⑥}{b_{2,2}} \\ a_{2,1} \cdot \overset{③}{b_{1,1}} + a_{2,2} \cdot \overset{④}{b_{2,1}} & a_{2,1} \cdot \overset{⑦}{b_{2,1}} + a_{2,2} \cdot \overset{⑧}{b_{2,2}} \end{bmatrix}$$

$$= \begin{bmatrix} AB_{1,1} & AB_{1,2} \\ AB_{2,1} & AB_{2,2} \end{bmatrix}$$

$m \times n \times k$

$= 2 \times 2 \times 2 = 8$ multiplications

$$(AB)C = \begin{bmatrix} AB_{1,1} \overset{①}{\cdot} C_{1,1} + AB_{1,2} \overset{②}{\cdot} C_{1,2} \\ AB_{2,1} \overset{③}{\cdot} C_{1,1} + AB_{2,2} \overset{④}{\cdot} C_{1,2} \end{bmatrix}$$

$2 \times 2 \times 1 = 4$ multiplications

$\therefore (AB)C$ requires $8 + 4 = 12$ multiplications

5.b) $A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}$ $B = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix}$ $C = \begin{bmatrix} c_{1,1} \\ c_{2,1} \end{bmatrix}$

$m \times n$ 2×2 $n \times k$ 2×2 $k \times p$ 2×1

$$BC = \begin{bmatrix} b_{1,1} \overset{1}{\cdot} c_{1,1} + b_{1,2} \overset{2}{\cdot} c_{2,1} \\ b_{2,1} \overset{5}{\cdot} c_{1,1} + b_{2,2} \overset{4}{\cdot} c_{2,1} \end{bmatrix}$$

$$= \begin{bmatrix} BC_{1,1} \\ BC_{2,1} \end{bmatrix}$$

$n \times k \times p$
$= 2 \times 2 \times 1$
$= 4$ multiplications

$$A(BC) = \begin{bmatrix} a_{1,1} \overset{1}{\cdot} BC_{1,1} + a_{1,2} \overset{2}{\cdot} BC_{2,1} \\ a_{2,1} \overset{3}{\cdot} BC_{1,1} + a_{2,2} \overset{4}{\cdot} BC_{2,1} \end{bmatrix}$$

$m \times n \times p = 2 \times 2 \times 1 = 4$ multiplications

c) $A = 4 \times 2$ $B = 2 \times 4$ $C = 4 \times 2$

$m \times n$ $n \times k$ $k \times p$

$$BC = \begin{matrix} 2+2+3+2 & 1+2+3+3 \\ 2+1+1+4 & 1+1+1+6 \end{matrix} = \begin{bmatrix} 9 & 9 \\ 8 & 9 \end{bmatrix}$$

$$A(BC) = \begin{matrix} 9+16 & 9+18 \\ 9+8 & 9+9 \\ 18+24 & 18+27 \\ 9+8 & 9+9 \end{matrix} = \begin{bmatrix} 25 & 27 \\ 17 & 18 \\ 42 & 45 \\ 17 & 18 \end{bmatrix}$$